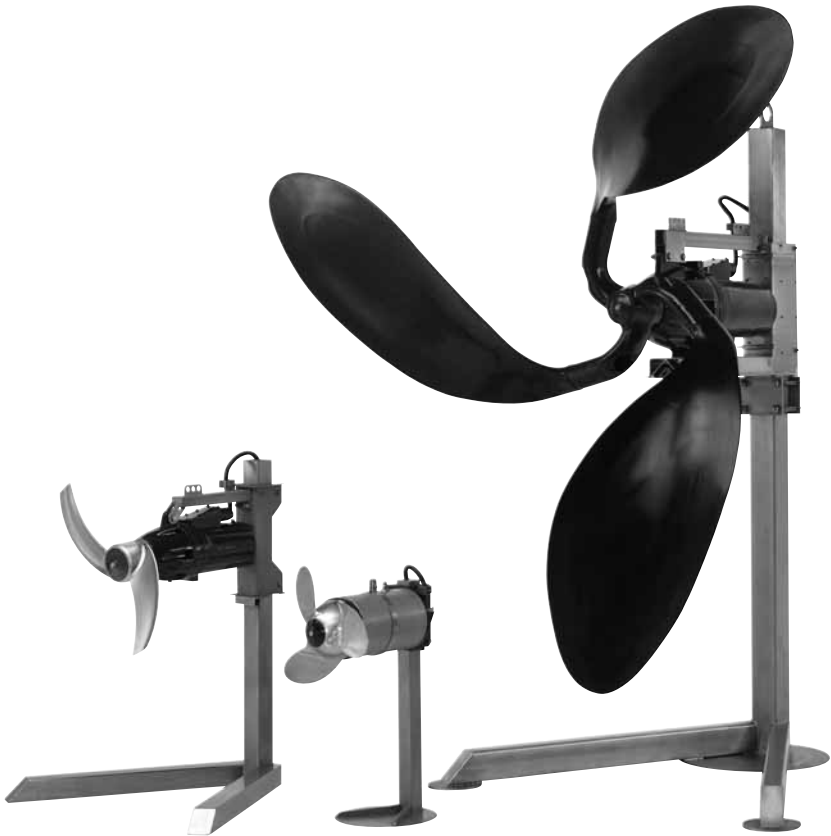


AMD, AMG, AFG Mixers and flowmakers

60 Hz

Installation and operating instructions



AMD, AMG, AFG

Mixers and flowmakers

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LIMITED WARRANTY

Products manufactured by GRUNDFOS PUMPS CORPORATION (Grundfos) are warranted to the original user only to be free of defects in material and workmanship for a period of 24 months from date of installation, but not more than 30 months from date of manufacture. Grundfos' liability under this warranty shall be limited to repairing or replacing at Grundfos' option, without charge, F.O.B. Grundfos' factory or authorised service station, any product of Grundfos' manufacture. Grundfos will not be liable for any costs of removal, installation, transportation, or any other charges which may arise in connection with a warranty claim. Products which are sold but not manufactured by Grundfos are subject to the warranty provided by the manufacturer of said products and not by Grundfos' warranty. Grundfos will not be liable for damage or wear to products caused by abnormal operating conditions, accident, abuse, misuse, unauthorised alteration or repair, or if the product was not installed in accordance with Grundfos' printed installation and operating instructions.

To obtain service under this warranty, the defective product must be returned to the distributor or dealer of Grundfos' products from which it was purchased together with proof of purchase and installation date, failure date, and supporting installation data. Unless otherwise provided, the distributor or dealer will contact Grundfos or an authorised service station for instructions. Any defective product to be returned to Grundfos or a service station must be sent freight prepaid; documentation supporting the warranty claim and/or a Return Material Authorisation must be included if so instructed.

GRUNDFOS WILL NOT BE LIABLE FOR ANY INCIDENTAL OR CONSEQUENTIAL DAMAGES, LOSSES, OR EXPENSES ARISING FROM INSTALLATION, USE, OR ANY OTHER CAUSES. THERE ARE NO EXPRESS OR IMPLIED WARRANTIES, INCLUDING MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE, WHICH EXTEND BEYOND THOSE WARRANTIES DESCRIBED OR REFERRED TO ABOVE.

Some jurisdictions do not allow the exclusion or limitation of incidental or consequential damages and some jurisdictions do not allow limit actions on how long implied warranties may last. Therefore, the above limitations or exclusions may not apply to you. This warranty gives you specific legal rights and you may also have other rights which vary from jurisdiction to jurisdiction.

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**Warning**

If these safety instructions are not observed, it may result in personal injury!

**Caution**

If these safety instructions are not observed, it may result in malfunction or damage to the equipment!



Notes or instructions that make the job easier and ensure safe operation.

**Warning**

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

2. General description

This booklet includes instructions for installation, operation and maintenance of Grundfos AMD and AMG mixers and AFG flowmakers which are designed for applications involving the mixing, i.e. the homogenisation and suspension, of liquids of low to medium viscosity (≤ 500 mPas).

The mixers are fitted with motors of 1.5 to 18.5 kW.

The flowmakers are fitted with motors of 1.5 to 7.5 kW.

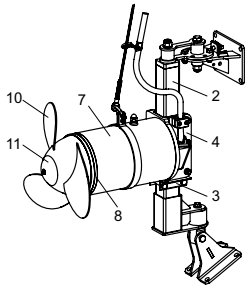


Fig. 1 AMD mixer

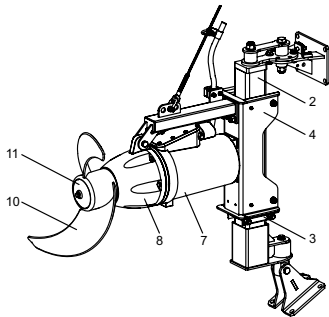


Fig. 2 AMG mixer

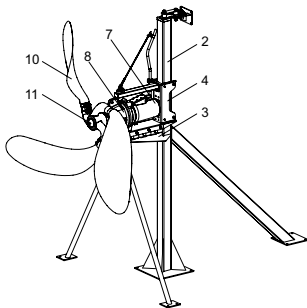


Fig. 3 AFG flowmaker

Pos.	Description
2	Column profile
3	Depth blocker
4	Motor bracket
7	Motor housing
8	AMD: Shaft seal housing AMG, AFG: Gearbox
10	Propeller
11	Hub

2.1 Applications

Grundfos mixers and flowmakers are designed for mixing applications in:

- municipal and industrial wastewater treatment
- industrial processes
- sludge treatment
- agriculture.

In order not to overload the mixers and flowmakers and expose them to corrosion, the following liquid limitations must be observed.

Liquid temperature range	5 to 40 °C
pH value	4 to 10
Maximum sludge index	125 ml/g
Maximum dynamic viscosity	500 mPas
Maximum density	1060 kg/m ³
Chloride content	≤ 200 mg/l (for stainless steel 1.4301)
Chloride content	≤ 1000 mg/l (for stainless steel 1.4404)

For the mixing of liquids with a dry solids content (DS) exceeding the values stated below, please contact Grundfos.

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TM04 2710 3210

TM04 2755 2908

2.1.1 Mixers

Mixers are suitable for applications involving sludge with a typical dry solids content as stated in the table below. Mixers are also suitable for a wide range of other applications involving similar liquids such as slurry and paper pulp.

Activated sludge	0.5 % DS
Selector zones	0.5 % DS
Anoxic zones	0.5 % DS
Bivalent zones	0.5 % DS
Anaerobic zones	0.5 % DS
Primary sludge	≤ 3 % DS
Secondary sludge	≤ 6 % DS
Digested sludge	≤ 8 % DS
Wastewater tank without screen	≤ 2 % DS
Wastewater tank with sand	≤ 2 % DS

2.1.2 Flowmakers

Flowmakers are suitable for activated sludge with a typical dry solids content of 0.5 to 1.0 % and for other liquids with a dry solids content of maximum 1.5 %.

2.2 Technical data

Voltage tolerance	3 x 380 V - 5 %/+ 10 %, 60 Hz
	3 x 440-480 V - 10 %/+ 10 %, 60 Hz
Enclosure class	IP68
Insulation class	F
Maximum installation depth	20 m below liquid level
Maximum number of starts per hour	20
Supply cable length	10, 15 or 25 m
Wire length on all winches	10, 15 or 25 m

The mixers and flowmakers are designed for continuous operation when fully submerged (S1 operation).

3. Identification

3.1 Type key

Code	Example	A	M	G.	15.	40.	394.	6.	0K.
A	Type range AMD, AMG, AFG								
M	Version Mixer								
F	Flowmaker								
G	Drive Gear-driven								
D	Direct-driven								
15	Motor output power, P2 Code from type designation/10 [kW] 1.5 kW								
40	Propeller diameter 40 cm								
[-]	Applications All applications								
B	Biology*								
394	Propeller speed 394 min ⁻¹								
[-]	Explosion protection Non-explosion-proof								
Ex	Explosion-proof								
5	Frequency 50 Hz								
6	60 Hz								
0K	Supply voltage and starting method 380 V, DOL								
1K	380 V, Y/D								
0P	440-480 V, DOL								
1P	440-480 V, Y/D								
0Z	Special, DOL								
1Z	Special, Y/D								
[-]	Generation First generation								
A	Second generation								
B	Third generation								

* Only for liquids with a dry solids content ≤ 1.5 %.

3.2 Nameplate

All mixers and flowmakers can be identified by means of the nameplate on the motor housing. The details on the nameplate are required for ordering of spare parts.

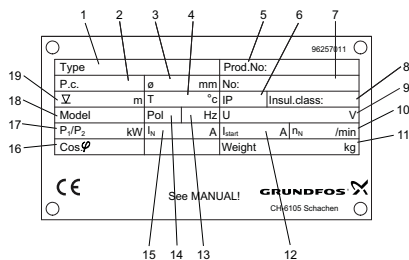


Fig. 4 Nameplate of AMD, AMG, AFG

Pos.	Description
1	Type designation
2	Production code
3	Propeller diameter
4	Liquid temperature range
5	Product number
6	Enclosure class according to IEC
7	Serial number
8	Insulation class
9	Rated voltage
10	Rated speed (propeller)
11	Weight
12	Starting current
13	Frequency
14	Number of poles
15	Rated current
16	Power factor
17	Motor power, P_1/P_2
18	Model
19	Maximum installation depth

The additional nameplate supplied with the mixer or flowmaker should be fixed in a visible position at the installation site.

4. Safety

4.1 General safety instructions

Warning



Before starting any work on mixers or flowmakers, make sure that the fuses have been removed or the mains switch has been switched off. It must be ensured that the power supply cannot be accidentally switched on.

Warning



These safety instructions as well as the instructions in each individual section must be followed when transporting, storing, handling and operating the mixer or flowmaker.

The mixer or flowmaker must be installed, connected, started up and serviced by qualified persons.

Beware of rotating parts.

It must be ensured that persons cannot accidentally fall into the tank, e.g. by installing a cover or railing.

5. Transportation, delivery and storage

5.1 Transportation

The individual components of the mixer or flowmaker must be packed carefully to prevent any damage to the surface protection during transportation.



Warning
Make sure that the mixer or flowmaker cannot roll or fall over.



Warning
Before attempting to lift or otherwise handle the individual components of the mixer or flowmaker, observe any local regulations that set limits for the weight of the components to be lifted manually by individuals, i.e. handled without the use of lifting equipment.

All lifting equipment must be rated for the purpose and checked for damage before any attempts to lift the components are made. The lifting equipment rating must under no circumstances be exceeded.

5.2 Delivery



Warning
A damaged mixer or flowmaker must not be installed.

On delivery, the mixer or flowmaker and any accessories supplied with it should be checked for transport damage. This also applies when the equipment is delivered to the installation site.

If the mixer, flowmaker or any accessories have been damaged, contact and inform your local Grundfos company before continuing to install the equipment. Do not dismantle a damaged new component for further inspection, unless instructed by your local Grundfos company.

The packaging material must be disposed of according to local regulations.

5.3 Storage

Mixers or flowmakers must be stored in a dry location in which the temperature is not subject to major fluctuations.

If the mixer or flowmaker has been stored for more than one year, the gearbox oil must be changed. The oil must be changed even if the unit has never been in use. This is necessary because of natural aging of mineral oil lubricants.

6. Installation

During installation, the mixer or flowmaker must only be lifted using the suspension point.

The lifting equipment supplied with the unit as well as the wire used for lifting and lowering the mixer or flowmaker into the tank must not be used as universal lifting equipment.

Caution
Never hang the mixer or flowmaker by the supply cable. Never let the mixer or flowmaker run while hanging in the lifting equipment.

See pages 131 and 132 for details about weights.

6.1 Positioning

Correct positioning of mixers and flowmakers is essential to ensure trouble-free operation and long life.

The guidelines in the following sections must be observed.

6.1.1 Mixers

The mixer should be submerged as deeply as possible.

- The mixer should be placed in such a way that a good mixing of the liquid in the entire tank is obtained. If more mixers are installed in the same tank, the mixers must not generate opposite flows.
- The distance between the propeller tip and the bottom of the tank, H_{MIN} in fig. 5, should be 20 to 40 cm.
- The distance from the propeller tip to the liquid surface, H_{ABOVE} in fig. 5, should be at least half of the propeller diameter.
- The distance between the propeller tip and the wall behind the mixer, L_{MIN} in fig. 5, must be at least 1.5 times the propeller diameter.

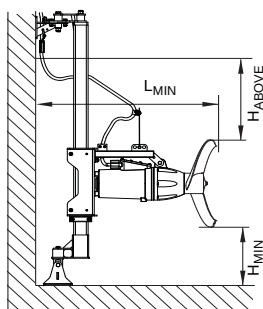


Fig. 5 Positioning sketch for mixers

6.1.2 Flowmakers

The flowmaker should be submerged as deeply as possible.

- The distance between the propeller tip and the bottom of the tank, H_{MIN} in fig. 6, should be 40 to 50 cm.
- The distance from the propeller tip to the liquid surface, H_{ABOVE} in fig. 6, should be at least 0.75 times the propeller diameter.
- The distance between the propeller tip and the wall behind the flowmaker, L_{MIN} in fig. 6, must be at least twice the propeller diameter.
- The sideways distance between the propeller tip and the tank wall must be at least 0.5 metres.
- If more flowmakers are installed in parallel, the distance between their propellers must be larger than half the propeller diameter.
- The distance to tank bends should be at least twice the propeller diameter.
- The distance from the propeller to aeration areas should be at least twice the propeller diameter.

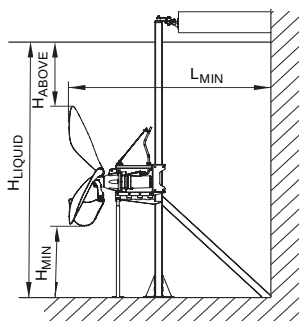


Fig. 6 Positioning sketch for flowmakers

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6.2 Installation instructions

6.2.1 Torques

All nuts and screws used for the installation must be made of stainless steel.

Use grease (Alu-paste) together with a spring washer or lock nut; otherwise, use Loctite or a similar product for lubrication and locking.

All stainless-steel nuts and screws should be tightened to the following torques:

	Screws F-Class 70 [Nm]	Screws F-Class 80 [Nm]
M6	8.8	11.8
M8	21.4	28.7
M10	44	58
M12	74	100
M16	183	245
M20	370	494

When mounting the hub/propeller, use lock washer (pos. 1107), and tighten the two shaft nuts (pos. 1106) to a torque of 50 Nm. See fig. 7.

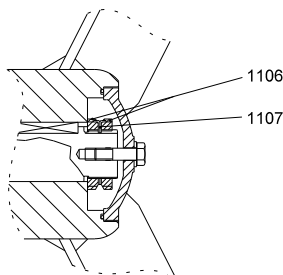


Fig. 7 Shaft nuts on propeller shaft

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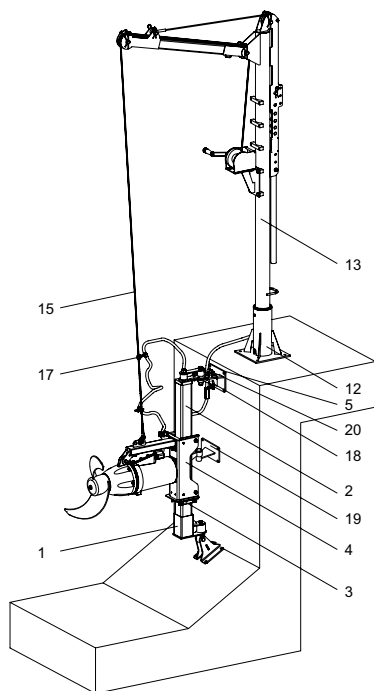


Fig. 8 Installation of AMD and AMG mixers

Position numbers in figures 8 and 9

Pos.	Description
1	Bottom fixation bracket
2	Column profile
3	Depth blocker
4	Motor bracket
5	Top fixation bracket incl. safety wire
12	Crane foot
13	Crane with winch
15	Lifting wire incl. shackle and wire clamp
17	Cable clamp
18	Cable sock incl. shackle
19	Intermediate fixation bracket
20	Wire clamping
31	Hole for fastening of safety wire

Procedure

Torques appear from section 6.2.1 *Torques*.

1. Drill the holes for the mounting screws for the top fixation bracket in the concrete.
2. Mount the screws, and fit the top fixation bracket.
3. Place the bottom fixation bracket in the right position (in vertical line to top fixation).
The bottom fixation bracket can be mounted in any angle from vertical to horizontal position.
4. Drill the holes for the mounting screws for the bottom fixation bracket.
5. Mount the screws, and fit the bottom fixation bracket.
6. Depending on the length of the column profile, weld the turnable part of an intermediate fixation bracket to the column profile.
7. Position and align the column profile in the bottom fixation bracket. Shorten the column profile (pos. 2) to the correct length to match the position of the insulator (pos. C) at the top fixation bracket. A gap of 5 to 10 mm between collar of insulator and column profile is optimal. See fig. 9.

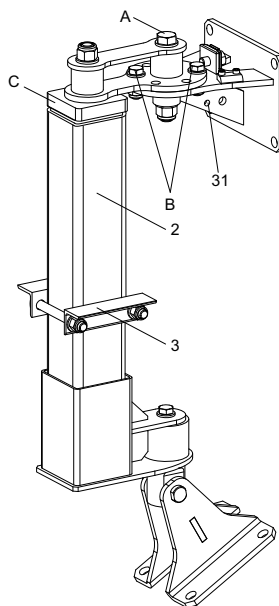


Fig. 9 Fixation

8. Remove the insulator and the turnable metal part by removing the centre screw (pos. A) and the two fixation screws (pos. B).
9. Adapt the outside of the square insulator to the inside of the column profile. The insulator must fit tightly inside the column profile.
10. Insert the column profile into the bottom fixation bracket, and mount the top end with the insulator and the turnable metal part on the already installed top fixation bracket. Tighten the three screws (pos. A and B) in the desired position. It is possible to adjust the angle in steps of 7.5° .
11. Fit an intermediate fixation bracket to the turnable part welded on to the column profile in step 6. Drill holes in the tank wall, fit screws in the bracket and tighten the screws.

Caution

Make sure that the mixer cannot be turned so much that the propeller touches the tank wall.

12. Fasten the depth blocker (pos. 3) in the right position.
13. Drill the holes for the mounting screws for the crane foot in the concrete.
14. Mount the crane foot, and fit and tighten the screws.
15. Mount the lifting wire (pos. 15) on the motor bracket using the shackle. See fig. 8.
16. Mount the top end of the safety wire to the hole (pos. 31) of the top fixation bracket using a shackle. The other end of the safety wire ends in a shackle through which the lifting wire must run.

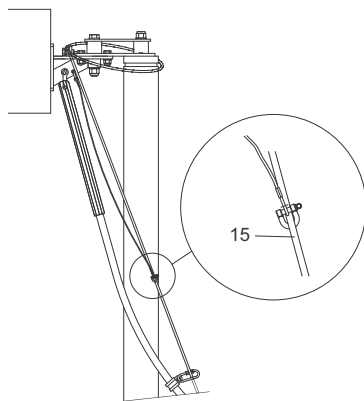


Fig. 10 Safety wire

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17. Fix the power supply cable to the lifting wire by means of a cable clamp approx. 0.8 m above the mixer. This will prevent the cable from falling down and becoming entangled in the propeller during operation. Connect the cable clamp to the lifting wire above the wire clamp by means of a snap hook. See fig. 11. Attach the supply cable to the lifting wire by means of cable clamps placed at 1 m intervals.

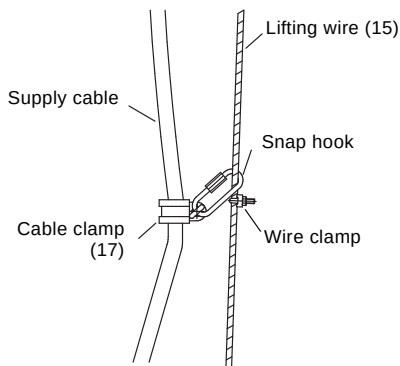


Fig. 11 Attaching the supply cable to the lifting wire

18. Position the crane in the foot, and mount the lifting wire in the drum of the winch.

Caution

At least three turns of wire must always remain on the drum. Otherwise the wire may break loose from the drum fixation.

Note

Consider and follow the separate installation and operating instructions for cranes.

19. Lift the complete mixer (motor bracket with motor) using the crane and slide it over the column profile.
20. Slowly lower the mixer into the tank and down to position on the depth blocker.
21. Mount the cable sock (pos. 18) to the top fixation bracket using the shackle, and pull the motor cable through it to the desired position. See fig. 12. The supply cable should be slightly tightened.

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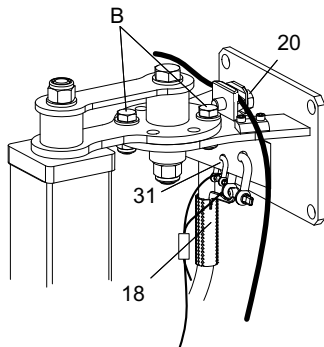


Fig. 12 Top fixation bracket with lifting and safety wires and cable sock

Caution

Dismantle the lifting wire from the crane before starting the mixer.

22. Dismantle the lifting wire from the winch and fit it to the wire clamping (pos. 20) on the top fixation bracket.

Use the lifting wire as a relief for the supply cable. For this reason, the lifting wire must always be tightened.

23. Check the distance between the propeller and the wall and the propeller and the bottom of the tank when the motor bracket is resting on the depth blocker. The mixer must under no circumstances touch other installations, bottom or wall. This also applies when the mixer is turned.



Warning

Make sure that the power supply to the control cabinet has been switched off.

24. Connect the supply cable to the terminals in the control cabinet.

6.2.3 Flowmakers

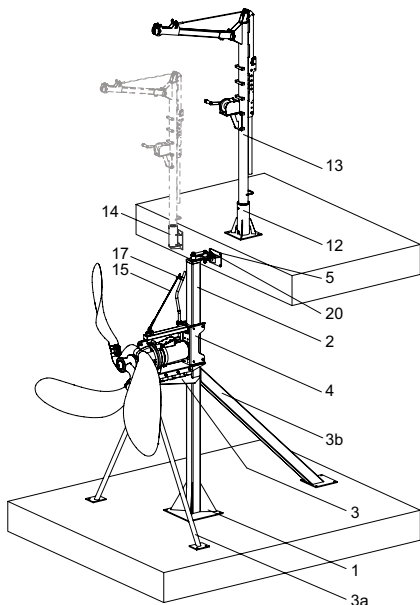


Fig. 13 Installation of AFG flowmaker

Position numbers in figures 13 and 19

Pos.	Description
1	Bottom fixation plate
2	Column profile
3	Depth blocker
3a	Front support leg
3b	Back support leg
4	Motor bracket
5	Top fixation bracket incl. safety wire
12	Crane foot
13	Crane with winch
14	Crane foot for vertical installation
15	Lifting wire incl. shackle and wire clamp
17	Cable clamp
20	Wire clamping
31	Hole for fastening of safety wire

Procedure

Torques appear from section 6.2.1 *Torques*.

Note

The propeller blades can be mounted before installation of AFG.xx.180/230/260 or after the crane has been installed.

6.2.4 Mounting the propeller blades of AFG.xx.180/230

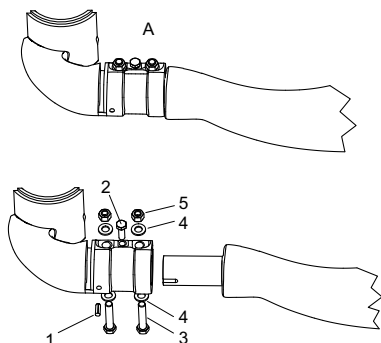


Fig. 14 AFG.xx.180/230

The fixation of the propeller blades is ensured by the pressure of the clamping jaws formed by the screws (pos. 3) and nuts (pos. 5). The function of the pin (pos. 1) is to position the blades in the right angle before tightening the clamping jaws.

Caution

1. Check the premounted pins (pos. 1) (only for positioning).
2. Tighten the screw (pos. 2) to widen the opening in the hub.
3. Insert the blade from above. Turn it slightly and let it rest on the pin. Turn back until the blade slides down and is flush with the hub.
4. Remove the screw (pos. 2).
5. Put Loctite 243 threadlocker on the thread of the screw (pos. 3).
6. Mount pos. 3, 4 and 5 in both holes and tighten by hand.
7. Tighten both nuts (pos. 5) with a torque wrench to 100 Nm (A4-80).
8. Carry out a visual check. See pos. A in fig. 14.
9. Refit the screw (pos. 2) and tighten only slightly.
10. Apply silicone to the joint between the blade and the hub.

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6.2.5 Mounting the propeller blades of AFG.xx.260

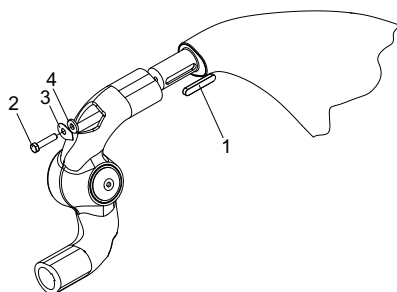


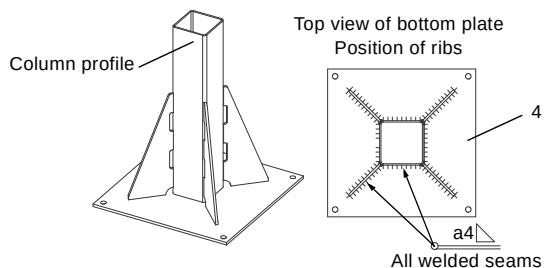
Fig. 15 AFG.xx.260

1. Drive the key (pos. 1) into the keyway.
2. Apply a little oil to the blade shaft and the hole in the hub.
3. First put the cover plate (pos. 3) and then the washer (pos. 4) on the screw (pos. 2).
4. Put Loctite 243 threadlocker on the thread of the screw (pos. 2).
5. Turn the hub, and insert the blade from above.
6. Turn the blade to correct position and let it slide into the hub.
7. Screw in screw (pos. 2) by hand and tighten it with a torque wrench to 183 Nm (A2-70).
8. Carry out a visual check.
9. Apply silicone to the joint between the blade and the hub and between cover plate (pos. 3) and the hub.

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6.2.6 All AFG flowmakers

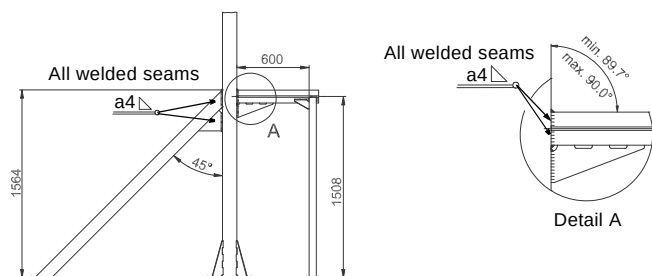
1. Weld the bottom fixation plate on the end of column profile in a workshop.



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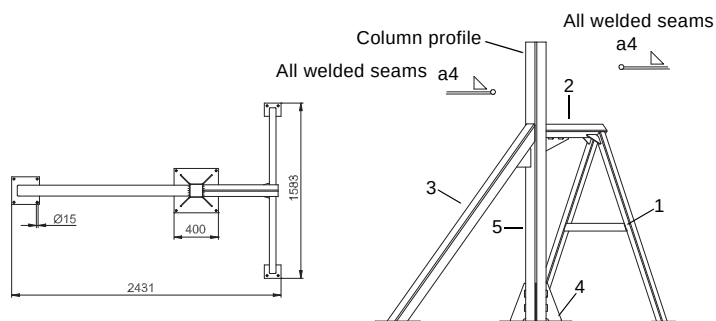
Fig. 16 Welding column profile on to bottom fixation plate

2. Place the depth blocker (pos. 2), the back support leg (pos. 3) and the front support legs (pos. 1) in the right position, and weld them on the column profile in a workshop. See position numbers in fig. 18.



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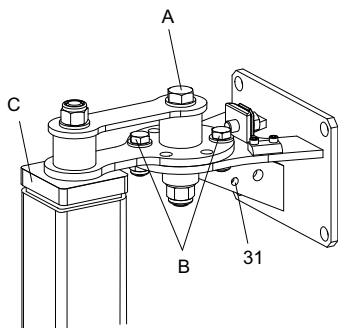
Fig. 17 Welding of column profile and support legs



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Fig. 18 Welding of column profile and support legs

3. Drill the holes for the mounting screws for the top fixation bracket in the concrete.
4. Mount the screws, and fit the top fixation bracket.

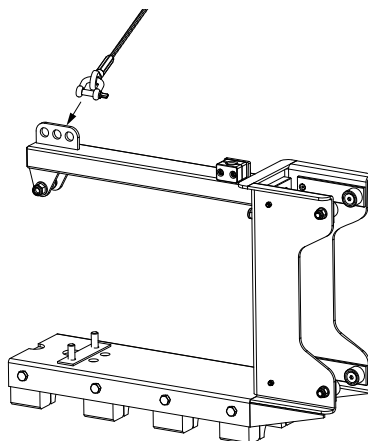


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Fig. 19 Top fixation

5. Shorten the column profile (pos. 5, fig. 18) to the correct length to match the position of insulator (pos. C) at the top fixation bracket. A gap of 5 to 10 mm between collar of insulator and column profile is optimal. See fig. 19.
6. Remove the insulator and the turnable metal part from the top fixation bracket by removing the centre screw (pos. A) and the two fixation screws (pos. B).
7. Adapt the outside of the square insulator to the inside of the column profile. The insulator must fit tightly inside the column profile.
8. Fit the turnable metal part (now fitted on top of the column profile) to the already mounted top fixation bracket. Tighten the three screws (pos. A and B) in the desired position. It is possible to adjust the angle in steps of 7.5 °.
9. Drill the holes for the mounting screws for the bottom fixation plate in the bottom of the tank, and insert the screws.
10. Tighten the screws in the bottom fixation plate.
11. Drill holes, mount and tighten screws of front and back support legs as well.
12. Drill the holes for the mounting screws for the crane foot in the concrete.
13. Mount the crane foot, and fit and tighten the screws.

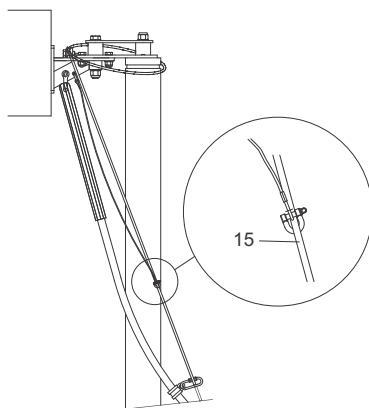
14. Mount the lifting wire (pos. 15, fig. 13) on the motor bracket using the shackle. See fig. 20.



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Fig. 20 Mounting of lifting wire on motor bracket

15. Mount the top end of the safety wire to the hole (pos. 31, fig. 19) of the top fixation bracket using a shackle. The other end of the safety wire ends in a shackle through which the lifting wire must run. See pos. 15 in fig. 21.



TM04 3932 0409

Fig. 21 Safety wire

16. Fix the power supply cable to the lifting wire by means of a cable clamp (pos. 17, fig. 22) approx. 0.8 m above the flowmaker. This will prevent the cable from falling down and becoming entangled in the propeller during operation. Connect the cable clamp to the lifting wire above the wire clamp by means of a snap hook. See fig. 22. Attach the supply cable to the lifting wire by means of cable clamps placed at 1 m intervals.

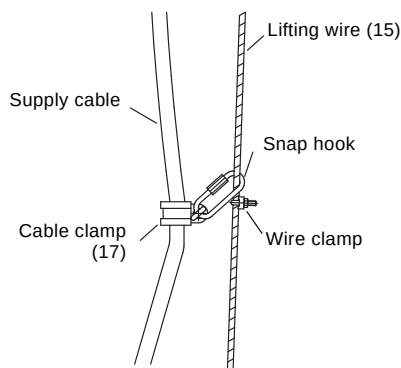


Fig. 22 Attaching the cable to the wire

17. Position the crane in the foot (pos. 12 or 14, fig. 13), and mount the lifting wire in the drum of the winch.

Caution

At least three turns of wire must always remain on the drum. Otherwise the wire may break loose from the drum fixation.

Note

Consider and follow the separate installation and operating instructions for cranes.

18. Lift the complete flowmaker (motor bracket with motor) using the crane and slide it over the column profile.
19. Slowly lower the flowmaker into the tank and down to position on the depth blocker.

20. Mount the cable sock (pos. 18) to the top fixation bracket using the shackle, and pull the motor cable through it, to the desired position. See fig. 23. The supply cable should be slightly tightened.

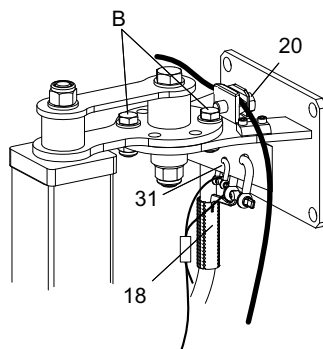


Fig. 23 Top fixation bracket with lifting and safety wires and cable sock

Caution

Dismantle the lifting wire from the crane before starting the flowmaker.

21. Dismantle the lifting wire from the winch and fit it to the wire clamping (pos. 20) on the top fixation bracket.

Use the lifting wire as a relief for the supply cable. For this reason, the lifting wire must always be tightened.

22. Check the distance between the propeller and the wall and the propeller and the bottom of the tank when the motor bracket is resting on the depth blocker. The flowmaker must under no circumstances touch other installations, bottom or wall. This also applies when the flowmaker is turned.



Warning

Make sure that the power supply to the control cabinet has been switched off.

23. Connect the power supply cable to the terminals in the control cabinet.

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7. Electrical connection

All electrical connections must be carried out by a qualified electrician in accordance with local regulations.

All currently valid national and local regulations relating to safety and accident prevention must be observed.

The mixer or flowmaker is designed for frequency converter operation in the 30 to 50 or 40 to 60 Hz range, depending on the rated frequency. If sufficiently cooled, however, the mixer or flowmaker can run at a minimum frequency of 10 Hz.

Warning



Before making any electrical connections, make sure that the fuses have been removed or the mains switch has been switched off. It must be ensured that the power supply cannot be accidentally switched on.

The supply voltage and frequency are marked on the mixer or flowmaker nameplate. Make sure that the mixer or flowmaker is suitable for the power supply available at the installation site.

The mixer or flowmaker is supplied complete with a supply cable of 10, 15 or 25 metres.

The motor is marked either with a Y (star) or a Δ (delta). Make this connection in an external control panel using conductors 1 to 6 of the supply cable.

Figure 24 shows a schematic drawing of these star and delta connections.

See also section 7.5 *Wiring diagrams*.

If the mixer or flowmaker is connected in delta during operation, the mixer or flowmaker can be started in star-delta starting.

Three-phase motors

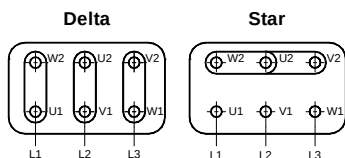


Fig. 24 Schematic drawing of delta and star connection

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7.1 Motor protection

The mixers and flowmakers are provided with the following type of motor protection:

Standard mixers incorporate three bimetallic PTO thermal switches (PTO = Protection Thermique à Ouverture). See fig. 27.

Standard flowmakers incorporate three PTC sensors (temperature sensors) according to DIN 44081. See fig. 28.

Function of thermal switches

The motor is protected against overheating by three thermal switches connected in series, one switch in each winding.

When the maximum winding temperature is reached, the switch will open the circuit and stop the motor.

When the windings have cooled to normal temperature, the switch will close the circuit and the motor can be restarted. Manual restarting is necessary.

See wiring diagram in fig. 27, section 7.5 *Wiring diagrams*.

Thermal switches (F6)

- Two conductors (terminals 11 and 12).
- Maximum operating voltage of switch: 250 V.
- Maximum switching current: 2.5 A at $\cos \phi = 1$.
- Cutting-out temperature: 150 °C.

Function of PTC sensors

When overheated, the motor will stop. Automatic restarting is not permitted in such cases.

This requires a thermistor trigger unit with a reconnection suppressor in the control circuit of the motor contactor.

See wiring diagram in fig. 28, section 7.5 *Wiring diagrams*.

Ø1, Ø2, Ø3: PTC sensors:

- Two conductors (terminals 31 and 32).
- Maximum voltage at the terminals:
 $U_{\max.} = 2.5 \text{ V (AC/DC)}$.
- Resistance between terminals 31 and 32:
 - at room temperature $R = 150 \text{ to } 750 \Omega$.
 - at cutting-out temperature (130 °C)
 $R \geq 4000 \Omega$.

Note

When the connection at terminals 31 and 32 is tested, the test voltage must not exceed 2.5 V (AC/DC). Use an ohmmeter for the test.

7.2 Gearbox/shaft seal housing protection

The gearbox/shaft seal housing is monitored for the ingress of water by means of a leak sensor incorporated in the gearbox/shaft seal housing.

If the monitoring function is required, the leak sensor must be connected to a Grundfos relay, type ALR-20/A. The ALR-20/A relay must be ordered separately. Part number: 96489569.

Note

The cable between the relay and the mixer or flowmaker must be maximum 50 m.

For longer distance, use an additional, screened cable. An external alarm indicator, if any, must be connected to the potential-free outputs, terminals 1 and 3 or 4, respectively.

Maximum load: 250 V, 5 A.

When the ALR-20/A relay is connected, a current of up to 10 mA will flow through the leak sensor (terminals 21 and 22). In case water penetrates into the oil chamber, the relay will trigger an alarm signal and/or switch off the motor.

See wiring diagram in fig. 27 or 28, section 7.5 *Wiring diagrams*.

Leak sensor

- Two conductors (terminals 21 and 22).
- Operating current: 1 to 10 mA.



Warning

When adjusting the relay, beware of electrical voltage.

To adjust the sensitivity of the ALR-20/A relay, proceed as follows:

1. Turn the adjusting screw on the relay **a** until the indicator light **b** lights up.
2. Turn the adjusting screw in the opposite direction until the indicator light goes out.
3. Continue to turn the adjusting screw 60 ° (same direction as under point 2).

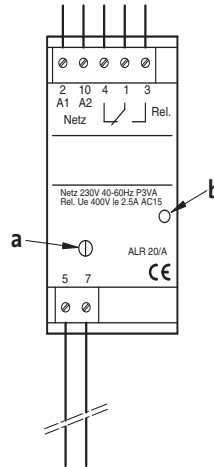


Fig. 25 Adjusting the relay



Fig. 26 ALR 20 relay connections

Note

Do not check the leak sensor with an ohmmeter or other measuring instruments. The leak sensor is an electronic component.

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7.3 Overload relays

The motor must be protected against overload via a thermal delay relay according to local regulations. The relay must be adjusted to the rated current stated on the nameplate.

In the case of star-delta starting, the adjustable value is to be $I_N \times 0.58$.

Incorporate electro-thermal all-pole triggers in all six mains conductors (U1, V1, W1 and U2, V2, W2).

7.4 Starting method

7.4.1 Mixers

Continuous operation

Motors of 1.5 to 2.2 kW can be started via direct starting.

We recommend star-delta starting, soft starter or frequency converter for motors of 3.0 kW and up.

Intermittent operation

We recommend star-delta starting, soft starter or frequency converter throughout the entire power range.

7.4.2 Flowmakers

Flowmakers must be started via a star-delta starting, soft starter or frequency converter.

7.5 Wiring diagrams

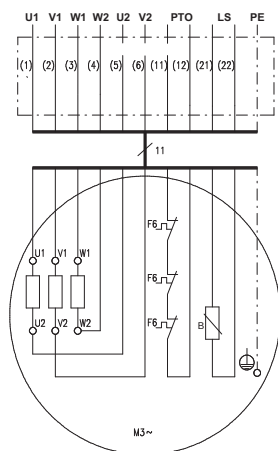


Fig. 27 Three thermal switches (PTO)

Terminals	Description
1, 2, 3, 4, 5, 6	Ends of the three stator windings (U1, U2, V1, V2, W1, W2)
11, 12	Thermal switches (F6)
21, 22	Leak sensor in gearbox (B)

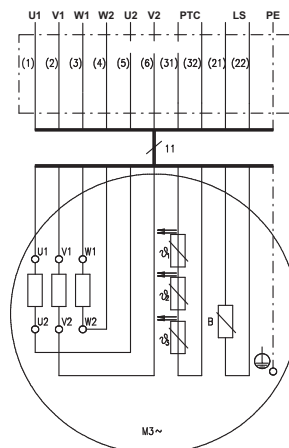


Fig. 28 Three PTC sensors

Terminals	Description
1, 2, 3, 4, 5, 6	Ends of the three stator windings (U1, U2, V1, V2, W1, W2)
31, 32	PTC sensors according to DIN 44081 (§1, §2, §3)
21, 22	Leak sensor in gearbox (B)

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7.6 Direction of rotation

When the electrical connections have been carried out, check that the mixer or flowmaker propeller is rotating in the correct direction. When viewed from the motor, the propeller should rotate clockwise. An arrow on the motor housing shows the correct direction of rotation.

If the mixer or flowmaker propeller rotates in the wrong direction, interchange two phases of the mains supply (L1, L2, L3).

7.7 Protection against electro-chemical corrosion

Two different metals or alloys cause electro-chemical corrosion if they are connected by an electrolyte. This applies if more than one mixer or flowmaker are installed in the same tank.

We recommend the following additional protection:

- galvanic separation of the earth conductor from the neutral conductor or
- galvanic separation of the mains supply by means of isolation transformer.

The earth conductor must be separated in such a way as to ensure that no direct current can flow through it. It must still function as a protective conductor. This can be achieved with a limiting unit (polarisation cell or anti-parallel diode) or an isolation transformer.

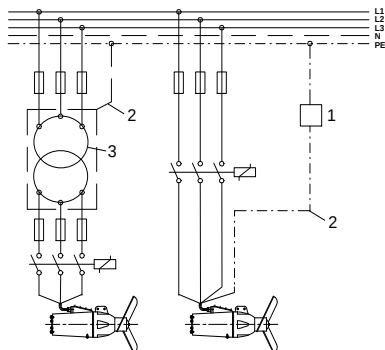


Fig. 29 Protection against electro-chemical corrosion

Pos.	Description
1	Limiting unit
2	Earth conductor
3	Isolation transformer



Warning

When using an isolation transformer, the ratio between starting current and rated current (I_A/I_N) must not be altered.

8. Start-up

Check the oil level in the gearbox/shaft seal housing before starting up the mixer or flowmaker. The oil level must cover between 50 and 75 % of the gearbox/shaft seal housing.



Warning

When slackening the oil level screw, note that pressure may have built up in the chamber. Do not remove the screw until the pressure has been fully relieved.

If required, fill oil into the gearbox/shaft seal housing through the oil filling hole (pos. 2). See fig. 30.

For oil quality and quantity, see section 9.3.1 Oil quality, gearbox and shaft seal housing.

It is necessary to remove the propeller of AMD to check the oil level.

If the mixer or flowmaker has been in stock for a period before start-up, see section 9.2 Service chart.

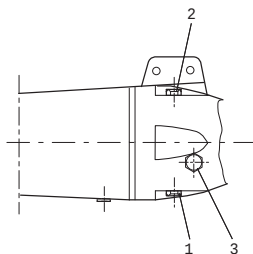


Fig. 30 Oil check and oil filling

Checks before start-up:

- Check that the propeller is rotating in the correct direction. See section 7.6 Direction of rotation.
- Make sure that the mixer or flowmaker is completely submerged in the liquid.

Caution

The mixer or flowmaker must always be submerged during operation.

- Make sure that there are no solid objects in the tank.



Warning

Make sure that no persons can fall into the tank.

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9. Service

Warning



Before starting any work on mixers or flowmakers, make sure that the fuses have been removed or the mains switch has been switched off. It must be ensured that the power supply cannot be accidentally switched on. All rotating parts must have stopped moving.

Before starting any work on mixers or flowmakers used in liquids which could constitute a hazard to health, thorough cleaning/venting of mixer or flowmaker, tank, etc. must be carried out according to local regulations.

9.1 Contaminated mixer or flowmaker

Caution

If a mixer or flowmaker has been used for a liquid which is injurious to health or toxic, the mixer or flowmaker will be classified as contaminated.

If Grundfos is requested to service the mixer or flowmaker, Grundfos must be contacted with details about the liquid, etc. *before* the mixer or flowmaker is returned for service. Otherwise Grundfos can refuse to accept the mixer or flowmaker for service.

Possible costs of returning the mixer or flowmaker are paid by the customer.

However, any application for service (no matter to whom it may be made) must include details about the liquid if the mixer or flowmaker has been used for liquids which are injurious to health or toxic.

9.2 Service chart

	Type	Service instructions	Lubrication	Inspection
Electric motor	All	Keep the motor housing clean (otherwise cooling is affected). The motor housing may only be dismantled by Grundfos.	The roller bearings are maintenance-free. They should be replaced if they become noisy.	For motors filled with dielectric oil, oil level check and oil change are not required.
Supply cable	All			The supply cable must be checked twice a year for surface damage, strain, kinks, etc. If damaged, the cable must be replaced by Grundfos.
Shaft seal housing	AMD		Change the oil after 8,000 operating hours. Minimum every two years. Check the lip seal and wear ring, and replace them in case of wear. If there is water in the oil, replace the mechanical shaft seal.	
Gearbox	AMG, AFG	Check the gearbox for leaks twice a year.	Change the oil after 500 operating hours. Then every 8,000 operating hours or after one year of operation. Check the lip seals, speedysleeve and wear ring, and replace them in case of wear. If there is water in the oil, replace the mechanical shaft seal.	Check the oil level twice a year. See section 9.3.3 <i>Oil quantity</i> . If refilling is required, see section 9.3.1 <i>Oil quality, gearbox and shaft seal housing</i> .
Propeller	All			Check the propeller blades regularly for wear and tear. Remove any material wound around the blades, such as ropes, threads, etc. which may cause uneven running and oscillation of the installation. In case of strong turbulence, cleaning is absolutely necessary.
Winch	All	Spray the winch with a protective coating of oil at regular intervals (to prevent corrosion).	The gear teeth and the bearing bushes must be lubricated twice a year with an all-purpose grease.	
Lifting wire	All	Regular oiling or greasing increases the life of the wire.		Check the wire regularly and always before using the winch. Replace the wire, if required.
Screws	All	Always check that all screws in the motor bracket are properly tightened. Check the screws in the bottom fixation bracket/ bottom fixation plate every time the tank is empty.		

9.3 Oil

9.3.1 Oil quality, gearbox and shaft seal housing

Gear oil designation for mixers (AMD, AMG) according to DIN 51502: ISO VG 68.
Flowmakers (AFG) according to ISO VG 220.

9.3.2 Oil quality, motor

Shell Fluid 4600.

The motor oil should only be changed if the motor is dismantled for service or repair.

9.3.3 Oil quantity

Mixers and flowmakers

Type	Gearbox/ shaft seal housing [l]	Motor [l]
AMD.25.45B.	0.25	-
AMD.30.45.		
AMD.35.45B.		
AMD.40.45.		
AMG.15.40.	1.2	2.6
AMG.22.52.		1.9
AMG.30.63.		
AMG.40.64.		
AMG.55.60.	2.5	5.9
AMG.75.68.		4.1
AMG.110.73.	2.5	
AMG.150.75.	4.0	6.8
AMG.185.75.		
AFG.14.130.	1.3	2.6
AFG.18.130.		1.9
AFG.30.130.		
AFG.40.130.		
AFG.13.180.	3.2	2.6
AFG.18.180.		1.9
AFG.27.180.		
AFG.37.180.		
AFG.15.230.	3.2	1.6
AFG.22.230.		1.9
AFG.30.230.		
AFG.40.230.		
AFG.55.260.	4.6	-
AFG.75.260.		

9.4 Oil change

AMD

To change the oil in the shaft seal housing, proceed as follows:

1. Remove the propeller.
 2. Remove the oil filling and oil drain screws.
 3. Drain the oil into a glass and leave it there for approx. 10 minutes, and check if it contains water.
- If the oil contains water, replace the shaft seal.

Note

Used oil must be disposed of in accordance with local regulations.

4. Fill the shaft seal housing with new oil.
See section 9.3.3 *Oil quantity*.
5. Fit the propeller.

AMG, AFG

To change the oil in the gearbox, proceed as follows:

1. Place the mixer or flowmaker in a horizontal position on supports, and place a pan underneath to collect the oil.

Warning



When slackening the oil level screw, note that pressure may have built up in the chamber. Do not remove the screw until the pressure has been fully relieved.

2. Slacken and remove the screw (pos. 2, fig. 31).
3. Slacken and remove the oil drain screw (pos. 1), and allow the oil to drain from the chamber into a glass. Leave the oil in the glass approx. 10 minutes, and check if it contains water. If the oil contains water, the shaft seal must be replaced.

Note

Used oil must be disposed of in accordance with local regulations.

4. Clean and refit the oil drain screw (pos. 1, fig. 31).
5. Fill oil into the oil chamber through the filling hole (pos. 2, fig. 31). The quantity of oil appears from section 9.3.3 and the quality from section 9.3.1.
6. Refit the screw (pos. 2).

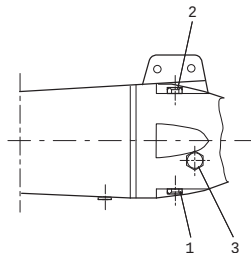


Fig. 31 Position of oil drain and oil filling screws

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10. Fault finding



Warning

Before starting any work on mixers or flowmakers, make sure that the fuses have been removed or the mains switch has been switched off. It must be ensured that the power supply cannot be accidentally switched on.

All rotating parts must have stopped moving.

10.1 Fault finding chart

Fault	Cause	Remedy
1. Mixer or flowmaker does not start.	a) No power supply or supply failure.	Call an electrician.
	b) Supply cable is faulty.	Call an electrician.
	c) Control system is faulty.	Call an electrician.
	d) Propeller not free to rotate.	Clean the propeller blades, and check manually that the propeller can rotate freely.
	e) Stator windings are faulty.	Contact Grundfos.
	f) Motor has cut out because of overheating.	Wait until the motor has cooled, and try to restart the mixer or flowmaker.
	g) Different phase voltages.	Call an electrician.
	h) Overload relay is set too low or is faulty.	Check the overload relay. Set the relay to the rated current. See section 7.3 <i>Overload relays</i> .
	i) Leak sensor has cut out the mixer or flowmaker.	Contact Grundfos.
	j) Humidity in motor.	Contact Grundfos.
2. Mixer or flowmaker starts but stops immediately.	a) Stator windings are faulty.	Contact Grundfos.
	b) Different phase voltages.	Call an electrician.
	c) Overload relay is set too low or is faulty.	Check the overload relay. Set the relay to the rated current. See section 7.3 <i>Overload relays</i> .
	d) Leak sensor has cut out the mixer or flowmaker.	Contact Grundfos.
	e) Humidity in motor.	Contact Grundfos.
3. No or inadequate circulation produced even if the motor is running.	a) Propeller rotates in the wrong direction.	Interchange two phases of the mains supply.
	b) Mixer or flowmaker runs on two phases.	• Check the electrical connections. • Replace faulty fuses. • Call an electrician.
	c) Internal parts are worn.	Contact Grundfos.
	d) Propeller blades are dirty or damaged.	Clean the blades, and inspect for any wear.
4. Mixer or flowmaker runs unevenly and is noisy.	a) Internal parts are worn.	Contact Grundfos.
	b) Propeller blades are dirty or damaged.	Clean the blades, and inspect for any wear.
	c) Faulty motor or gearbox roller bearings.	Contact Grundfos.
	d) Oscillations caused by the installation (resonance).	Check installation design.

Fault	Cause	Remedy
5. High current and power consumption.	a) Wrong voltage supply or supply failure.	Call an electrician.
	b) Supply cable is faulty.	Call an electrician.
	c) Control system is faulty.	Call an electrician.
	d) Propeller not free to rotate.	Clean the propeller blades, and check manually that the propeller can rotate freely.
	e) Stator windings are faulty.	Contact Grundfos.
	f) Mixer or flowmaker runs on two phases.	• Check the electrical connections. • Replace faulty fuses. • Call an electrician.
	g) Internal parts are worn.	Contact Grundfos.
	h) Faulty motor or gearbox roller bearings.	Contact Grundfos.

11. Technical data

11.1 Motor

AMD, AMG, AFG

Protective system	IP68 to 20 m
Insulation class	F
Sealing	Radial shaft seal ring/mechanical shaft seal
Material, motor housing	Cast iron, grade 25 (EN-GJL-250) AMD: stainless steel (AISI 316)

11.2 Gearbox

AMG, AFG

Type	Planetary gearbox
Gears	Hardened and ground steel
Seal monitoring	Leak sensor incorporated in gearbox
Drive end bearings	Two tapered roller bearings
Material, gear casing	Cast iron, grade 25 (EN-GJL-250)

11.3 Shaft seals

Sealing against ingress of surrounding liquid

AMD	V-ring, lip seal and mechanical shaft seal made of SiC/SiC
AMG, AFG	Two lip seals and one mechanical shaft seal made of tungsten carbide/ tungsten carbide or SiC/SiC

11.4 Propeller

AMD

Number of blades	3
Propeller diameter	450 mm
Propeller design	Self-cleaning, optimum-flow design
Material, propeller	Stainless steel
Material, hub	

AMG

Number of blades	2
Propeller diameter	400, 520, 600, 630, 640, 680, 730 and 750 mm
Material, propeller with hub	Stainless steel (1.4301)

AFG.xx.130

Number of blades	2
Propeller diameter	1340 mm
Material, propeller with hub	Ductile moulded polyamide with stainless-steel hub

AFG.xx.180, AFG.xx.230, AFG.xx.260

Number of blades	2: 1.3 / 1.4 / 1.5 / 1.8 / 2.2 / 2.7 / 3.7 / 5.5 kW 3: 3.0 / 4.0 / 7.5 kW
Propeller diameter	AFG.xx.180: 1800 mm AFG.xx.230: 2300 mm AFG.xx.260: 2600 mm
Material, hub	Cast iron (EN-GJS-400-15)
Material, propeller blades	Polyurethane (Baydur) with cast-iron reinforcement

11.5 Sound pressure level

The sound pressure level of the mixer or flowmaker is lower than 70 dB(A).

12. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

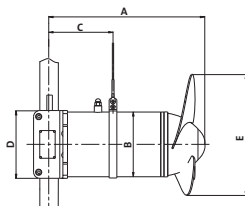
1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

Subject to alterations.

Appendix

Dimensions and weights

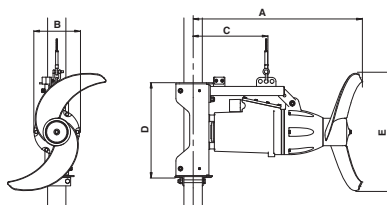
AMD



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Type	Rated power [kW]	A [mm]	B [mm]	C [mm]	D [mm]	E [mm]	Weight incl. motor bracket and 10 m cable [kg]
AMD.25.45B.	2.5	456	210	180	220	450	50
AMD.30.45.	3.0	491	210	200	220	450	59
AMD.35.45B.	3.5						
AMD.40.45.	4.0						

AMG

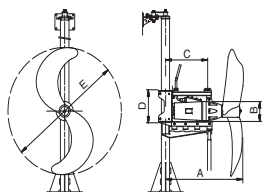


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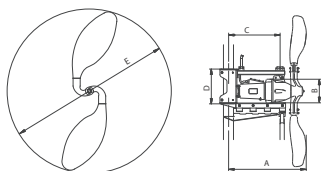
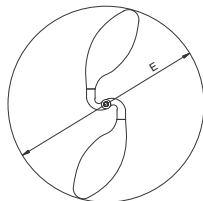
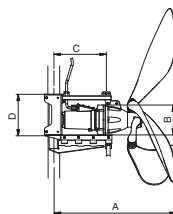
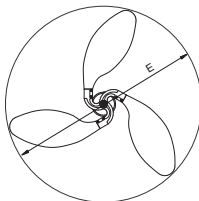
Type	Rated power [kW]	A [mm]	B [mm]	C [mm]	D [mm]	E [mm]	Weight incl. motor bracket and 10 m cable [kg]
AMG.15.40.	1.5	720	200	320	410	400	78
AMG.22.52.	2.2	735				520	78
AMG.30.63.	3.0	750				630	85
AMG.40.64.	4.0	760				640	85
AMG.55.60.	5.5	980	270	430	460	600	166
AMG.75.68.	7.5	990				680	168
AMG.110.73.	11.0	1010				730	177
AMG.150.75.	15.0	1160				750	275
AMG.185.75.	18.5	1180	315		460	750	280

AFG

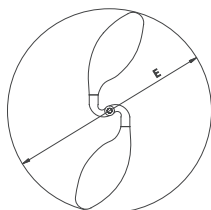
AFG.xx.130.xx



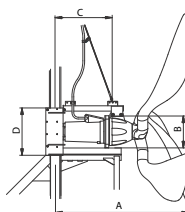
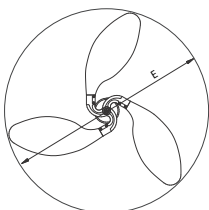
AFG.xx.180.xx

AFG.15.230
AFG.22.230AFG.30.230
AFG.40.230

AFG.55.260



AFG.75.260



Type	Rated power [kW]	A [mm]	B [mm]	C [mm]	D [mm]	E [mm]	Weight incl. motor bracket and 10 m cable [kg]
AFG.14.130.	1.4	855	200	353	400	1340	100
AFG.18.130.	1.8						100
AFG.30.130.	3.0						105
AFG.40.130.	4.0						105
AFG.13.180.	1.3	1200	302	570	400	1800	190
AFG.18.180.	1.8						198
AFG.27.180.	2.7						
AFG.37.180.	3.7						
AFG.15.230.	1.5	1200	302	570	400	2300	200
AFG.22.230.	2.2						200
AFG.30.230.	3.0						233
AFG.40.230.	4.0						233
AFG.55.260.	5.5	1403	315	640	535	2600	345
AFG.75.260.	7.5	1500	315	680	535	2660	391

TM02 6346 0305 - TM02 6345 0305 - TM04 3959 0409 - TM04 3957 0409

Argentina

Bombas GRUNDFOS de Argentina S.A.
Ruta Panamericana km. 37.500 Lote 34A
1619 - Garin
Pcia. de Buenos Aires
Phone: +54-3327 414 444
Telefax: +54-3327 411 111

Australia

GRUNDFOS Pumps Pty. Ltd.
P.O. Box 2040
Regency Park
South Australia 5942
Phone: +61-8-8461-4611
Telefax: +61-8-8340 0155

Austria

GRUNDFOS Pumpen Vertrieb
Ges.m.b.H.
Grundfosstraße 2
A-5082 Grödig/Salzburg
Tel.: +43-6246-883-0
Telefax: +43-6246-883-30

Belgium

N.V. GRUNDFOS Bellux S.A.
Boomsesteenweg 81-83
B-2630 Aartselaar
Tél.: +32-3-870 7300
Télécopie: +32-3-870 7301

Belorussia

Представительство ГРУНДФОС в
Минске
220123, Минск,
ул. В. Хоружей, 22, оф. 1105
Тел.: +(37517) 233 97 65,
Факс: +(37517) 233 97 69
E-mail: grundfos_minsk@mail.ru

Bosnia/Herzegovina

GRUNDFOS Sarajevo
Trg Heroja 16,
BiH-71000 Sarajevo
Phone: +387 33 713 290
Telefax: +387 33 659 079
e-mail: grundfos@bih.net.ba

Brazil

BOMBAS GRUNDFOS DO BRASIL
Av. Humberto de Alencar Castelo
Branco, 630
CEP 09850 - 300
São Bernardo do Campo - SP
Phone: +55-11 4393 5533
Telefax: +55-11 4343 5015

Bulgaria

Grundfos Bulgaria EOOD
Slatina District
Iztocna Tangenta street no. 100
BG - 1592 Sofia
Tel. +359 2 49 22 200
Fax: +359 2 49 22 201
email: bulgaria@grundfos.bg

Canada

GRUNDFOS Canada Inc.
2941 Brighton Road
Oakville, Ontario
L6H 6C9
Phone: +1-905 829 9533
Telefax: +1-905 829 9512

China

GRUNDFOS Pumps (Shanghai) Co. Ltd.
50/F Maxdo Center No. 8 XingYi Rd.
Hongqiao development Zone
Shanghai 200336
PRC
Phone: +86-021-612 252 22
Telefax: +86-021-612 253 33

Croatia

GRUNDFOS CROATIA d.o.o.
Cebini 37, Buzin
HR-10010 Zagreb
Phone: +385 1 6595 400
Telefax: +385 1 6595 499
www.grundfos.hr

Czech Republic

GRUNDFOS s.r.o.
Čajkovského 21
779 00 Olomouc
Phone: +420-585-716 111
Telefax: +420-585-716 299

Denmark

GRUNDFOS DK A/S
Martin Bachs Vej 3
DK-8850 Bjerringbro
Tlf.: +45-87 50 50 50
Telefax: +45-87 50 51 51
E-mail: info_GDK@grundfos.com
www.grundfos.com/DK

Estonia

GRUNDFOS Pumps Eesti OÜ
Peterburi tee 92G
11415 Tallinn
Tel: + 372 606 1690
Fax: + 372 606 1691

Finland

OY GRUNDFOS Pumput AB
Mestarintie 11
FIN-01730 Vantaa
Phone: +358-3066 5650
Telefax: +358-3066 56550

France

Pompes GRUNDFOS Distribution S.A.
Parc d'Activités de Chesnes
57, rue de Malacombe
F-38290 St. Quentin Fallavier (Lyon)
Tél.: +33-4 74 82 15 15
Télécopie: +33-4 74 94 10 51

Germany

GRUNDFOS GMBH
Schlüterstr. 33
40699 Erkrath
Tel.: +49-(0) 211 929 69-0
Telefax: +49-(0) 211 929 69-3799
e-mail: infoservice@grundfos.de
Service in Deutschland:
e-mail: kundendienst@grundfos.de

Greece

GRUNDFOS Hellas A.E.B.E.
20th km. Athinon-Markopoulou Av.
P.O. Box 71
GR-19002 Peania
Phone: +0030-210-66 83 400
Telefax: +0030-210-66 46 273

Hong Kong

GRUNDFOS Pumps (Hong Kong) Ltd.
Unit 1, Ground floor
Siu Wai Industrial Centre
29-33 Wing Hong Street &
68 King Lam Street, Cheung Sha Wan
Kowloon
Phone: +852-27861706 / 27861741
Telefax: +852-27858664

Hungary

GRUNDFOS Hungária Kft.
Park u. 8
H-2045 Törökbálint,
Phone: +36-23 511 110
Telefax: +36-23 511 111

India

GRUNDFOS Pumps India Private Limited
118 Old Mahabalipuram Road
Thoraiakkam
Chennai 600 096
Phone: +91-44 2496 6800

Indonesia

PT GRUNDFOS Pompa
Jl. Rawa Sumur III, Blok III / CC-1
Kawasan Industri, Puloagung
Jakarta 13930
Phone: +62-21-460 6909
Telefax: +62-21-460 6910 / 460 6901

Ireland

GRUNDFOS (Ireland) Ltd.
Unit A, Merrywell Business Park
Ballymount Road Lower
Dublin 12
Phone: +353-1-4089 800
Telefax: +353-1-4089 830

Italy

GRUNDFOS Pompe Italia S.r.l.
Via Gran Sasso 4
I-20060 Truccazzano (Milano)
Tel.: +39-02-95838112
Telefax: +39-02-95309290 / 95838461

Japan

GRUNDFOS Pumps K.K.
Gotanda Metalion Bldg., 5F,
5-21-15, Higashi-gotanda
Shiagawa-ku, Tokyo
141-0022 Japan
Phone: +81 35 448 1391
Telefax: +81 35 448 9619

Korea

GRUNDFOS Pumps Korea Ltd.
6th Floor, Aju Building 679-5
Yeoksam-dong, Kangnam-ku, 135-916
Seoul, Korea
Phone: +82-2-5317 600
Telefax: +82-2-5633 725

Latvia

SIA GRUNDFOS Pumps Latvia
Deglava biznesa centrs
Augusta Deglava iela 60, LV-1035, Rīga,
Tālr.: + 371 714 9640, 7 149 641
Fakss: + 371 914 9646

Lithuania

GRUNDFOS Pumps UAB
Smolensko g. 6
LT-03201 Vilnius
Tel: + 370 52 395 430
Fax: + 370 52 395 431

Malaysia

GRUNDFOS Pumps Sdn. Bhd.
7 Jalan Peguam U1/25
Glenmarie Industrial Park
40150 Shah Alam
Selangor
Phone: +60-3-5569 2922
Telefax: +60-3-5569 2866

México

Bombas GRUNDFOS de México S.A. de C.V.
Boulevard TLC No. 15
Parque Industrial Stiva Aeropuerto
Apodaca, N.L. 66600
Phone: +52-81-8144 4000
Telefax: +52-81-8144 4010

Netherlands

GRUNDFOS Netherlands
Veluwezoom 35
1326 AE Almere
Postbus 22015
1302 CA ALMERE
Tel.: +31-88-478 6336
Telefax: +31-88-478 6332
E-mail: info_gnl@grundfos.com

New Zealand

GRUNDFOS Pumps NZ Ltd.
17 Beatrice Tinsley Crescent
North Harbour Industrial Estate
Albany, Auckland
Phone: +64-9-415 3240
Telefax: +64-9-415 3250

Norway

GRUNDFOS Pumper A/S
Strømsveien 344
Postboks 235, Leirdal
N-1011 Oslo
Tlf.: +47-22 90 47 00
Telefax: +47-22 32 21 50

Poland

GRUNDFOS Pompy Sp. z o.o.
ul. Klonowa 23
Baranowo k. Poznań
PL-62-081 Przechmirowo
Tel: (+48-61) 650 13 00
Fax: (+48-61) 650 13 50

Portugal

Bombas GRUNDFOS Portugal, S.A.
Rua Calvet de Magalhães, 241
Apartado 1079
P-2770-153 Paço de Arcos
Tel.: +351-21-440 76 00
Telefax: +351-21-440 76 90

România

GRUNDFOS Pompe România SRL
Bd. Biruintei, nr 103
Pantelimon county Ilfov
Phone: +40 21 200 4100
Telefax: +40 21 200 4101
E-mail: romania@grundfos.ro

Russia

ООО Грундфос
Россия, 109544 Москва, ул. Школьная 39
Тел. (+7) 495 737 30 00, 564 88 00
Факс (+7) 495 737 75 36, 564 88 11
E-mail grundfos.moscow@grundfos.com

Serbia

GRUNDFOS Predstavništvo Beograd
Dr. Milutina Ivkovića 2a/29
YU-11000 Beograd
Phone: +381 11 26 47 877 / 11 26 47 496
Telefax: +381 11 26 48 340

Singapore

GRUNDFOS (Singapore) Pte. Ltd.
24 Tuas West Road
Jurong Town
Singapore 638381
Phone: +65-6865 1222
Telefax: +65-6861 8402

Slovenia

GRUNDFOS d.o.o.
Šlandrova 8b, SI-1231 Ljubljana-Črnuče
Phone: +386 1 568 0610
Telefax: +386 1 568 0619
E-mail: slovenia@grundfos.si

South Africa

Corner Mountjoy and George Allen Roads
Wilbart Ext. 2
Bedfordview 2008
Phone: (+27) 11 579 4800
Fax: (+27) 11 455 6066
E-mail: lsmart@grundfos.com

Spain

Bombas GRUNDFOS España S.A.
Camino de la Fuentecilla, s/n
E-28110 Algete (Madrid)
Tel.: +34-91-848 8800
Telefax: +34-91-628 0465

Sweden

GRUNDFOS AB
Box 333 (Lunnagårdsgatan 6)
431 24 Mölndal
Tel.: +46(0)771-32 23 00
Telefax: +46(0)31-331 94 60

Switzerland

GRUNDFOS Pumpen AG
Bruggacherstrasse 10
CH-8117 Fällanden/ZH
Tel.: +41-1-806 8111
Telefax: +41-1-806 8115

Taiwan

GRUNDFOS Pumps (Taiwan) Ltd.
7 Floor, 219 Min-Chuan Road
Taichung, Taiwan, R.O.C.
Phone: +886-4-2305 0868
Telefax: +886-4-2305 0878

Thailand

GRUNDFOS (Thailand) Ltd.
92 Chaloe Phrakiat Rama 9 Road,
Dokmai, Pravej, Bangkok 10250
Phone: +66-2-725 8999
Telefax: +66-2-725 8998

Turkey

GRUNDFOS POMPA San. ve Tic. Ltd. Sti.
Gebze Organize Sanayi Bölgesi
İhsan dede Caddesi,
2. yol 200. Sokak No. 204
41490 Gebze/ Kocaeli
Phone: +90 - 262-679 7979
Telefax: +90 - 262-679 7905
E-mail: satis@grundfos.com

Ukraine

ТОВ ГРУНДФОС УКРАЇНА
01010 Київ, Вул. Московська 86,
Тел.: (+38 044) 390 40 50
Факс: (+38 044) 390 40 59
E-mail: ukraine@grundfos.com

United Arab Emirates

GRUNDFOS Gulf Distribution
P.O. Box 16768
Jebel Ali Free Zone
Dubai
Phone: +971-4- 8815 166
Telefax: +971-4-8815 136

United Kingdom

GRUNDFOS Pumps Ltd.
Grovebury Road
Leighton Buzzard/Beds. LU7 8TL
Phone: +44-1525-850000
Telefax: +44-1525-850011

U.S.A.

GRUNDFOS Pumps Corporation
17100 West 118th Terrace
Olathe, Kansas 66061
Phone: +1-913-227-3400
Telefax: +1-913-227-3500

Uzbekistan

Представительство ГРУНДФОС в
Ташкенте
700000 Ташкент ул. Усмана Носира 1-й
тулик 5
Телефон: (3712) 55-68-15
Факс: (3712) 53-36-35

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Repl. 97761673 1110

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